

# The Epistemic Foundation of the Endogeneity Paradox: Common Knowledge, Intentionality Mismatch, and the Sequential Protocol

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## Abstract

Dochkina (2026) reports a striking empirical finding: among multi-agent LLM systems, a hybrid Sequential protocol, in which agent ordering is fixed but role selection is fully endogenous, outperforms both centralized coordination (+14%,  $p < 0.001$ ) and fully autonomous protocols (+44%, Cohen's  $d = 1.86$ ,  $p < 0.0001$ ).

The paper names this the endogeneity paradox but does not offer a theoretical mechanism explaining why completed outputs from predecessors are informationally superior to intentions, historical patterns, or a central plan. This response supplies that mechanism.

Drawing on the Generalized Intentionality Mismatch Theorem (Lerer, 2025a), the theory of common knowledge as an epistemic device (Pinker, 2025; Aumann, 1976), and the Extended Phenotype Theory applied to institutional design (Lerer, 2025b), this paper argues that the Sequential protocol succeeds because it is the minimal institutional structure capable of generating common knowledge among agents with heterogeneous intentionality, without requiring those agents to model each other's mental states.

The result generalizes beyond LLM systems to any multi-agent coordination problem, including legal institutions, collegial tribunals, bicameral legislatures, and procedural sequences in both common law and civil law traditions, where the same epistemic condition governs equilibrium stability.

**Keywords:** *endogeneity paradox, common knowledge, intentionality mismatch, Sequential protocol, multi-agent LLM, extended phenotype, evolutionary game theory, heterogeneous intentionality, multilevel selection, bicameralism, institutional design.*

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## 1. The Unexplained Core of Dochkina's Result

Dochkina (2026) establishes, across 25,000 task runs, eight LLM models, four to 256 agents, and eight coordination protocols, that the Sequential protocol produces significantly superior outcomes compared to both the Coordinator (centralized, exogenous) and the Shared (fully

autonomous, endogenous) protocols. The explanation offered is informational: Sequential agents observe completed outputs of predecessors rather than intentions (Broadcast), historical patterns (Shared), or a single coordinator's plan (Coordinator). Completed outputs are, the paper argues, informationally superior.

This explanation is correct but incomplete. It identifies the what: completed outputs are better information. It does not explain the why. What makes completed outputs categorically different from intentions or historical patterns as coordination inputs? The answer requires a theoretical framework that Dochkina's paper, by design, does not develop. That framework is the theory of common knowledge as an epistemic device, combined with a formal account of why heterogeneous intentionality makes intention-based coordination structurally unstable.

The distinction matters practically. If we understand only that Sequential works, we can replicate it. If we understand why it works, we can generalize the principle to any coordination architecture, predict when it will fail, and design institutional structures beyond multi-agent LLM systems that exploit the same mechanism.

## 2. The Generalized Intentionality Mismatch Theorem

The Generalized Intentionality Mismatch Theorem (Lerer, 2025a) establishes that in any multi-agent system where agents hold heterogeneous intentional models, that is, where agents not only have different goals but different beliefs about each other's goals, different beliefs about those beliefs, and so on, no coordination protocol that requires agents to model each other's mental states can achieve stable equilibrium without a fixed-point condition on the belief hierarchy.

Formally: let  $A$  be a set of agents, each holding an intentional model  $I_i$  representing its beliefs about the goals and beliefs of all other agents. If the mapping  $I_i$  is sensitive to the full belief hierarchy (agents model each other modeling each other modeling each other...), then for heterogeneous  $I_i$ , the system has no guaranteed Nash equilibrium reachable by finite rounds of mutual modeling. Coordination requires either (a) homogenization of intentional models, which is computationally expensive and behaviorally unrealistic, or (b) a coordination device that makes intentional modeling irrelevant.

Dochkina's Coordinator protocol fails precisely because it requires one agent to model all others well enough to assign them roles. When that model is imperfect, and it always is under heterogeneous intentionality, the assignments are suboptimal and the other agents have no mechanism to correct them. The Shared protocol fails because it requires each agent to build its strategy on historical patterns that implicitly encode the past intentions of others, which may not generalize to the current task. Both protocols demand intentional modeling where none is reliable.

The Sequential protocol sidesteps the problem entirely. Each agent observes what predecessors actually did, not what they intended or what history suggests they might do. Completed outputs are intention-free: they are facts about the world, not mental states. An agent deciding its role after observing three completed contributions does not need to model what those three agents intended; it needs only to assess what has been done and what remains to be done. The

intentionality mismatch problem dissolves because the protocol replaces intentional modeling with factual observation.

### **3. Common Knowledge as the Missing Mechanism**

Pinker (2025), following Aumann (1976), draws the decisive distinction between mutual knowledge and common knowledge. Mutual knowledge is the condition in which A knows  $p$ , B knows  $p$ , A knows that B knows  $p$ , and so on through some finite number of iterations, but where the chain is not closed. Common knowledge is the limit of that chain: everyone knows  $p$ , everyone knows that everyone knows  $p$ , and this is common knowledge, all at once.

The difference is not merely formal. Rubinstein's (1989) email game shows that no finite chain of mutual confirmations is sufficient to guarantee coordination: if there is any positive probability of the chain being broken, rational agents will not act on it. Common knowledge, by contrast, is generated by a single self-evident public event, a promulgated law, a broadcast signal, a completed output visible to all subsequent agents, that collapses the infinite regress in one move.

The Sequential protocol is, in this precise sense, a common-knowledge-generating machine. When agent  $k$  completes its output and that output becomes visible to all subsequent agents, the following condition obtains: every subsequent agent knows what agent  $k$  did, every subsequent agent knows that every other subsequent agent knows what agent  $k$  did, and this is common knowledge within the system. No agent needs to query the intentions behind the output; the output itself is the self-evident public event.

The Coordinator protocol does not generate this condition. The coordinator's assignments are not self-evidently derived from completed work; they reflect one agent's intentional model, which other agents know only imperfectly. The Shared protocol does not generate it either: historical patterns are private observations that different agents weight differently, and there is no self-evident event that synchronizes their interpretations. Only Sequential produces, at each step, a public completed fact from which all subsequent agents can update simultaneously.

This is why Dochkina observes that Sequential provides agents with "the optimal information type: not intentions, not history, not someone else's plan, but completed outputs". The theoretical reason is that completed outputs are the only information type that generates common knowledge among agents with heterogeneous intentionality.

### **4. The Capability Threshold as an ESS Condition**

Dochkina reports a second finding that the paper does not fully explain: below a capability threshold, self-organization reverses and rigid structure becomes beneficial. Claude Sonnet 4.6 shows that autonomy helps (+3.5% in free-form vs. fixed-role), while GLM-5 shows that autonomy hurts (-9.6%). The threshold is characterized by three capabilities: self-reflection, deep reasoning, and instruction following.

The Evolutionary Game Theory (EGT) framework, as developed in Lerer (2025a) and Lerer (2025c), provides the natural interpretation. An Evolutionarily Stable Strategy (ESS) is a strategy that, once adopted by a population, cannot be invaded by a mutant strategy. The

Sequential protocol is an ESS only among agents whose capability profile satisfies the threshold conditions: they must be able to assess their own competence relative to what predecessors have done (self-reflection), chain multi-step inferences about what remains to be done (deep reasoning), and respect the protocol's structural constraint even when deviation might seem locally attractive (instruction following).

Below the threshold, the Sequential protocol is not an ESS. An agent incapable of self-reflection cannot determine when to abstain voluntarily; it either contributes suboptimally or abstains excessively. An agent incapable of deep reasoning cannot identify the gap left by predecessors. An agent that does not follow instructions introduces noise into the output chain that breaks the common-knowledge-generating property of Sequential. In all three cases, the protocol degrades below the fixed-structure alternative.

This is formally analogous to the spandrel problem identified in Gould and Lewontin (1979) and applied to institutional design in Lerer (2025b): a structure that is adaptive under one set of environmental conditions becomes a liability under another. The Sequential protocol is not universally superior; it is superior conditional on the capability profile of the agents inhabiting it. Prescribing Sequential to systems below the threshold is the institutional equivalent of applying a coordination norm to a population that lacks the cognitive prerequisites for that norm to function as a focal point.

## **5. Voluntary Self-Abstention and the Deontic Power of Sequential**

Among the emergent properties Dochkina documents, voluntary self-abstention is the most theoretically significant. In the Sequential protocol, 38 out of 60 non-contributing agents withdrew by their own decision (endogenous), compared to zero voluntary withdrawals in the Coordinator protocol (all exclusions were coordinator-directed). Claude Sonnet 4.6 shows an 8.6% voluntary abstention rate; GLM-5 shows 0.8%.

This asymmetry has a precise interpretation in Searle's (1995) terms, as applied to institutional analysis in Lerer (2025b): the Sequential protocol generates deontic power, the capacity to create normative facts from brute facts through collective assignment of functions. When an agent observes that three predecessors have covered a domain completely, the normative conclusion that its contribution in that domain would be redundant emerges from the observation itself, without any external authority declaring it. The protocol assigns, implicitly, the function "abstain when the task is covered" to the agent's self-assessment, and the agent executes that function because the common knowledge of what has been done makes the normative conclusion self-evident.

The Coordinator protocol cannot generate this condition because the deontic structure is externally imposed: the coordinator decides who abstains, not the agents themselves. An agent excluded by the coordinator does not develop the capacity for self-abstention; it learns only that external authority determines participation. The Shared protocol cannot generate it because the absence of real-time visibility means agents cannot assess coverage; they act on stale priors and produce redundant contributions.

The practical consequence, confirmed empirically by Dochkina, is that voluntary self-abstention correlates with higher quality ( $Q = 0.875$  for Sequential vs.  $Q = 0.767$  for Coordinator). The theoretical explanation is that self-abstention is the behavioral signature of a system that has internalized the common knowledge of task coverage: agents abstain because they know, and know that all others know, that the domain is covered. That epistemic condition, not any external instruction, generates the coordination efficiency.

## 6. Dynamic Role Invention as Memetic Radiation

Dochkina reports that the Role Stability Index (RSI) converges to zero: with eight agents, 5,006 unique role names were observed; with 64 agents, 5,010 (+0.1%). Agents do not consolidate into fixed positions but reinvent their specialization for each task.

The memetic framework developed in Lerer (2025b) provides the natural vocabulary. A legal meme, or in this context a role meme, is the minimum unit of functional specialization capable of replicating within a system. The convergence of RSI to zero does not mean there are no role memes; it means the system has undergone memetic radiation: a rapid diversification of role variants, each adapted to the specific task context, analogous to the adaptive radiation that follows the opening of a new ecological niche.

The ecological niche, in this case, is the slot left open by the outputs of predecessor agents in each specific task. Because each task generates a different sequence of predecessor outputs, and because the common knowledge of those outputs is the input to role selection, the effective niche space is as large as the space of possible task-output combinations. RSI converges to zero not because roles are random but because the fitness landscape for roles is maximally task-specific: the same functional specialization that is optimal after output sequence A is suboptimal after output sequence B.

This finding challenges a foundational assumption in organizational design theory, as Dochkina correctly notes: that roles should be pre-assigned because role stability reduces coordination costs. The EGT interpretation is that pre-assigned roles are an ESS only in environments where the task distribution is sufficiently stable that the fitness landscape for roles does not change significantly across tasks. In high-variance task environments, exactly the L3 and L4 adversarial tasks where Sequential shows its largest advantage, pre-assignment is an anti-pattern because it fixes the phenotype to a landscape that no longer exists.

## 7. Generalization: Collegial Tribunals, Bicameral Legislatures, and Sequential Law

The most significant theoretical contribution of Dochkina's result is not specific to LLM systems. The endogeneity paradox is a general property of any multi-agent coordination system where agents have heterogeneous intentionality and where the task requires complementary rather than additive contributions.

The collegial tribunal is among the oldest institutional implementations of the Sequential protocol, and it appears in every legal tradition. When judges on a panel circulate draft opinions sequentially, each subsequent judge reads the completed votes of predecessors before drafting their own contribution. The mechanism that generates common knowledge here is not stare

decisis, which operates vertically across tribunals and across time, but the circulation of the projected vote itself: each judge observes a completed output, identifies the argumentative gap it leaves open, and contributes accordingly. The fact that collegial courts universally adopt an odd number of members, three, five, seven, or nine depending on jurisdiction and historical period, is the institutional solution to the aggregation problem at the voting stage. But the Sequential protocol operates upstream of that vote, in the drafting phase, where each judge reads what predecessors have written before composing their own opinion. The Intentionality Mismatch Theorem predicts that this protocol produces more stable and internally consistent reasoning than either a single rapporteur who distributes arguments to colleagues (the Coordinator equivalent) or simultaneous independent drafting without circulation (the Shared equivalent).

Bicameralism extends the Sequential logic to a second order. The reviewing chamber observes the complete output of the originating chamber before deliberating: it does not model the intentions of the lower house but responds to its enacted text. This is precisely the epistemic structure that Washington, in a conversation attributed to him by Jefferson, described through an enduring metaphor: just as coffee is poured into a saucer to cool it, legislation is sent to the Senate to be cooled. The metaphor captures the common-knowledge effect of Sequential at the inter-chamber level. The reviewing chamber, having before it a completed legislative output, can identify internal contradictions, gaps, and conflicts with existing law that are invisible to a chamber deliberating without that prior output. The Sequential protocol thus operates at two nested levels simultaneously: within each chamber (sequential amendment, successive readings, committee circulation) and between chambers (the originating output as the fixed input to the reviewing deliberation).

This nesting is not incidental. It suggests that Sequential protocols are composable structures: a Sequential protocol at level  $n$  can serve as the completed-output input to a Sequential protocol at level  $n+1$ . The institutional wisdom embedded in bicameralism is that a single pass of Sequential, however well-designed, does not exhaust the coordination gains available; a second pass operating on the output of the first generates a further layer of common knowledge that the first pass cannot produce. The implications for multi-agent LLM systems are explored in section 9.

The adversarial structure of civil and criminal proceedings follows the same logic at yet another level of abstraction. One party presents its case completely before the other responds. The responding party does not model its adversary's intentions; it responds to its adversary's completed output. This structure is universal: it appears in the alegatos and contestacion of civil law procedure, in the written memorials exchanged before oral argument, and in the sequence of charges and defenses that procedural codes in every jurisdiction impose. The common knowledge of what the first party has argued is the self-evident public event that collapses the infinite regress of mutual modeling and allows the adjudicator to evaluate contributions without having to reconstruct the intentions behind them.

The Extended Phenotype framework (Lerer, 2025b; Dawkins, 1982) identifies these institutional structures as phenotypic expressions of the underlying cognitive architecture: the Sequential protocol is not a peculiarity of any legal tradition but an emergent property of the

human capacity for common-knowledge-based coordination, expressed wherever multi-agent decision-making requires complementary rather than additive contributions.

## **8. The Constitutional Framework: Ring Theory, Multilevel Selection, and the ESS of Governance**

Dochkina proposes a three-ring constitutional framework for autonomous AI organizations: Ring 1 (immutable, human-only: mission and values), Ring 2 (standards: human plus system), Ring 3 (protocols: fully autonomous). The principle is that the closer to "why," the more human control; the closer to "how," the more system autonomy.

This framework is correct in direction but incomplete in its theoretical grounding. The ring structure is not arbitrary: it maps onto the multilevel selection framework developed in Wilson and Sober (1994) and integrated with the Extended Phenotype program in Lerer (2025b). That integration does not require any tension between Dawkins and Gould. Gould does not deny that selection operates at the level of the gene or the organism; his argument is that the modern Darwinian synthesis must account for the fact that constraints operating at higher levels, developmental, structural, and phylogenetic, shape and limit what selection at lower levels can produce. These constraints are real selective pressures, not merely filters. The three-ring structure reflects exactly this logic: Ring 1 defines the fitness criterion and the structural constraints within which lower-level variation is evaluated; Ring 2 sets group-level standards that constrain individual-level optimization; Ring 3 is the domain of individual-level protocol variation, fully endogenous but bounded by the constraints flowing downward from the rings above.

The ESS condition for this governance structure requires that the fitness criteria at each level be consistent: that what maximizes individual-level fitness (Ring 3) also contributes to group-level fitness (Ring 2), which also contributes to system-level fitness (Ring 1). When these conditions are met, the three-ring structure is self-sustaining without external enforcement. When they are not, when Ring 3 autonomy allows individual agents to optimize locally in ways that degrade group performance, the structure collapses. This is the Gouldian point applied to governance design: the adaptive value of Ring 3 autonomy is conditional on the constraints imposed by Rings 1 and 2. Remove those constraints and autonomy becomes a liability, exactly as unconstrained variation becomes a liability when developmental or structural constraints are absent in biological systems.

The empirical confirmation in Dochkina is the capability threshold finding: below-threshold models show that Ring 3 autonomy degrades Ring 2 performance. The theoretical interpretation is that the ESS condition for the three-ring structure requires agents whose cognitive architecture supports alignment between individual optimization (role selection) and collective optimization (task coverage). Where that alignment fails, Ring 3 must be contracted toward Ring 2, which is exactly what the reversal from Sequential to fixed-structure represents in below-threshold models.

## **9. The Open Problem: Adversarial Tasks, Nested Sequential, and the Limits of Single-Pass Coordination**

Dochkina documents a sharp quality degradation at L4 adversarial tasks: 37.7% drop from L1 (Cohen's  $d = 22.9$ ), with  $Q = 0.614$  for the best protocol at L4. The paper frames this as a "current frontier" without a theoretical account of why adversarial tasks are categorically harder.

The Intentionality Mismatch Theorem provides the account. L4 tasks involve conflicting stakeholder interests, incomplete information, and no single correct answer. In the game-theoretic vocabulary, L4 tasks are not coordination games but mixed-motive games: the agents' objectives are not merely different but partially opposed. The Sequential protocol generates common knowledge of completed outputs, but in mixed-motive games, common knowledge of an adversary's output does not simplify coordination. It may complicate it. If agent  $k$ 's output reveals a commitment that conflicts with another agent's objective, the common knowledge of that conflict does not create a focal point; it creates a standoff.

The theoretical prediction is that the Sequential protocol's advantage over Coordinator will shrink as task adversariality increases, and that the optimal protocol for L4 tasks may require a different epistemic structure: one that generates common knowledge not of completed outputs but of the conflict structure itself. This is the institutional equivalent of the adversarial proceeding, where the common knowledge of opposing positions is the input to a third-party decision-maker rather than to the agents themselves.

The bicameralism analysis in section 7 opens a more specific research direction. If Sequential protocols are composable, that is, if a second Sequential pass operating on the output of a first can generate a further layer of common knowledge unavailable to a single pass, then nested Sequential architectures may outperform flat Sequential on L4 tasks precisely because the second pass operates on the conflict structure revealed by the first rather than on the original task. The first pass produces outputs that make the mixed-motive structure of the task visible; the second pass coordinates around that structure rather than around the task directly. This is formally analogous to what a reviewing chamber does with contentious legislation: it does not re-deliberate the original policy question but coordinates around the conflict structure that the originating chamber's output has made explicit.

This yields a testable prediction absent from Dochkina's experimental design: a two-pass Sequential architecture, in which a second group of agents observes the full output of the first group and coordinates around it, should show smaller quality degradation on L4 tasks than single-pass Sequential, while showing no improvement on L1 and L2 tasks where a single pass already achieves near-ceiling performance. The prediction follows from the common-knowledge framework and does not require any modification to the underlying agents or models.

The research agenda this opens is precise: what protocol generates, in a multi-agent system with partially opposed objectives, the common knowledge of the conflict structure that allows a resolution without requiring the agents to model each other's intentions? Dochkina's data showing Mission Relevance 4.00/4.00 for both Claude and DeepSeek on Sequential L3 tasks is consistent with the hypothesis that a shared superordinate mission can function as the focal point that resolves adversarial conflicts, though L4 data is not disaggregated in the paper.



## 10. Implications for Institutional Design Beyond LLM Systems

The generalization of Dochkina's result to institutional design yields prescriptions that differ from both standard organizational theory and standard legal theory.

The first prescription is that sequential visibility of completed work, not role assignment or hierarchical authority, is the fundamental coordination device for systems with heterogeneous intentionality. Institutional designers who invest in role clarity and hierarchical definition are optimizing the wrong variable. What matters is the epistemic accessibility of what has been done, not the normative specification of who should do what.

The second prescription is that voluntary abstention is a signal of epistemic health, not a symptom of agent failure. An institution in which participants freely withdraw from decisions outside their competence, without being told to do so by any authority, has achieved the common knowledge of task coverage that Sequential generates. An institution in which participation is always mandatory, or always determined by external authority, has not.

The third prescription, derived from the capability threshold finding, is that the Sequential principle cannot be imposed on populations below the cognitive threshold that makes it viable. Legal reforms that replicate the structural form of Sequential institutions (sequential procedure, public outputs, transparent deliberation) without ensuring that participants have the self-reflection capacity to use them correctly will reproduce the Coordinator or Shared failure modes in institutional drag. The form of the protocol is not sufficient; the epistemic prerequisites of the participants must be met.

The fourth prescription, derived from the bicameralism analysis and the nested Sequential prediction, is that single-pass Sequential may be insufficient for high-adversariality coordination problems. Institutional designers facing multi-stakeholder conflicts with partially opposed interests should consider architectures in which a second deliberative body coordinates around the output of the first rather than around the original dispute. This is not a novel institutional invention: it is the theoretical justification for reviewing chambers, appellate courts, and constitutional review bodies that has been implicit in institutional practice for centuries. The framework developed here makes that justification explicit and connects it to a measurable prediction about coordination quality.

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*AI assistance disclosure: An AI language model was used as a research support tool for locating sources, extracting information from the literature, and reviewing drafts. The thesis, theoretical framework, arguments, and conclusions of this paper are the sole intellectual contribution of the author. This paper represents independent research.*